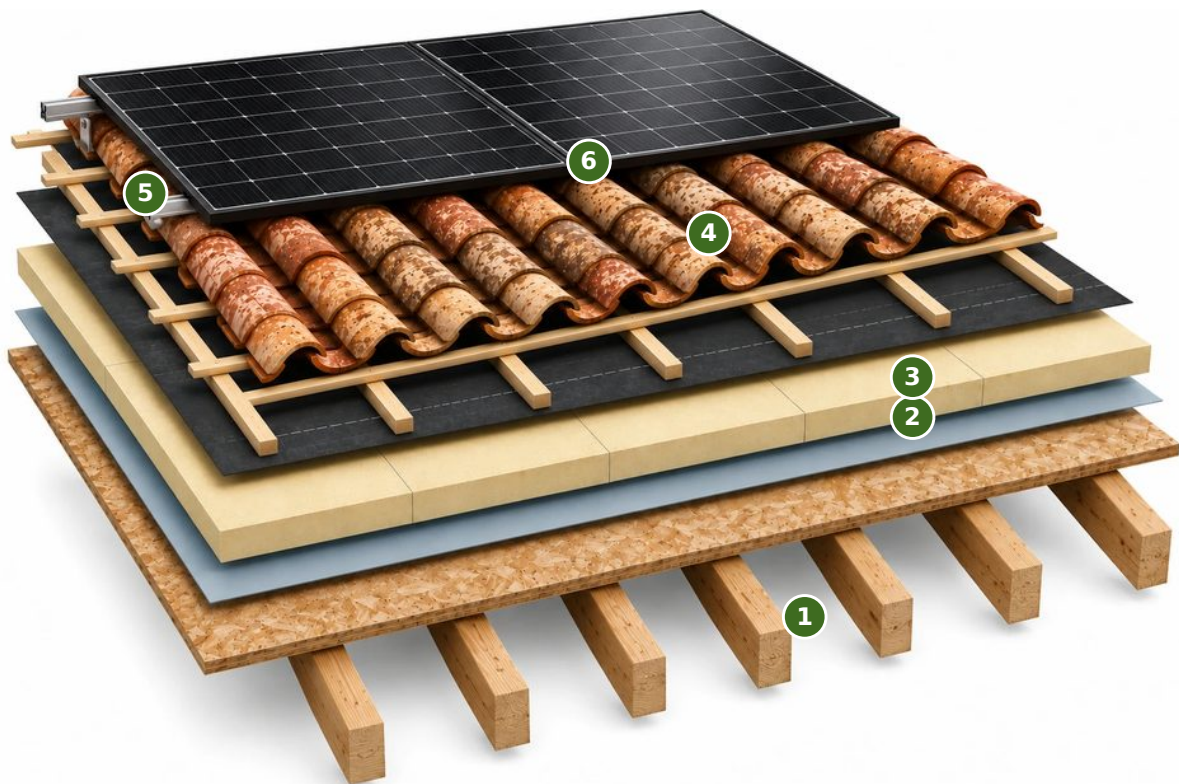




SOLAR PANELS & BATTERY SYSTEM

Roof integration · Mounting details · Electrical system

SOLAR ON A TRADITIONAL TILED ROOF



Illustrative roof build-up; final fixing, weatherproofing and load design are project-specific.

ROOF BUILD-UP & SOLAR INTEGRATION

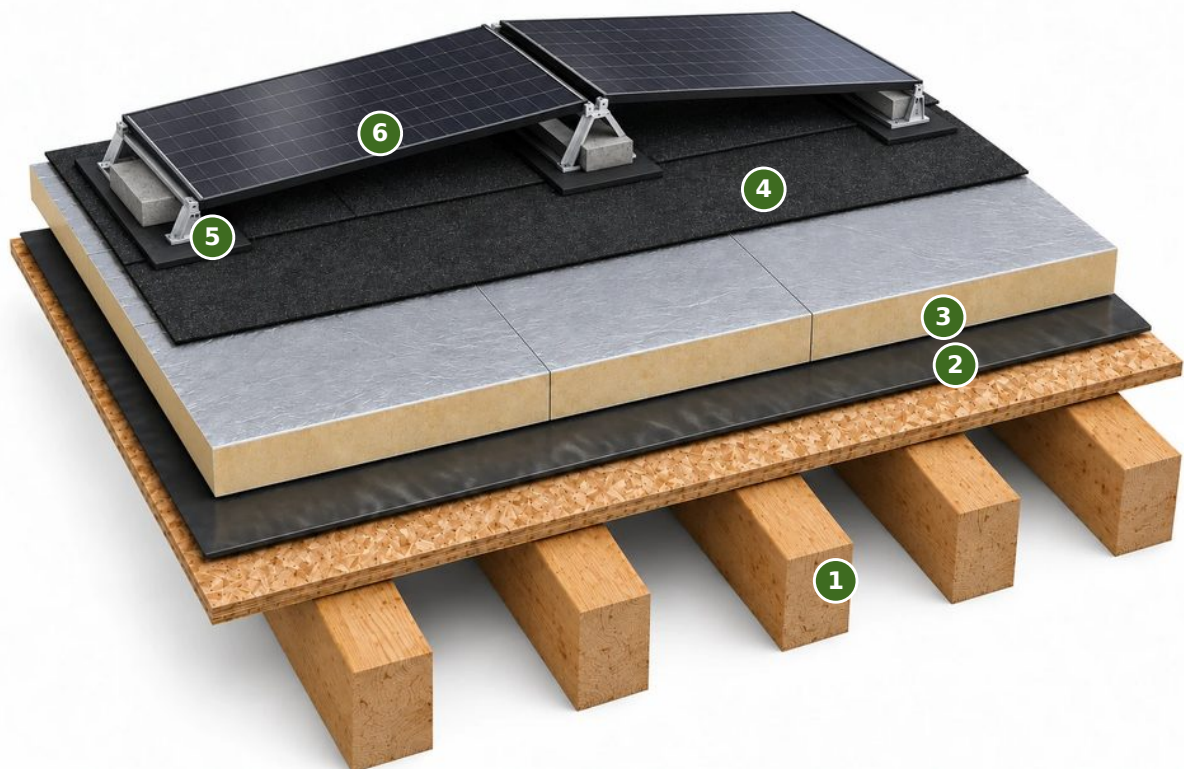
- | | |
|---------------------------------|---------------------------------|
| 1 Structural timber & roof deck | 4 Terracotta roof tiles |
| 2 Vapour control & insulation | 5 Structural roof hooks & rails |
| 3 Underlay & ventilated battens | 6 Photovoltaic modules |

Example insulated roof build-ups. Existing roofs may have a different construction; solar installation does not automatically include new insulation or a roof renovation.

SOLAR PANELS & BATTERY SYSTEM

Roof integration · Mounting details · Electrical system

SOLAR ABOVE AN INSULATED FLAT ROOF



Illustrative roof build-up; final fixing, weatherproofing and load design are project-specific.

ROOF BUILD-UP & SOLAR INTEGRATION

- | | |
|--|---|
| 1 Structural joists & roof deck | 4 Continuous waterproofing |
| 2 Vapour control layer | 5 Protection pads & support system |
| 3 Rigid thermal insulation | 6 Photovoltaic modules |

Example insulated roof build-ups. Existing roofs may have a different construction; solar installation does not automatically include new insulation or a roof renovation.

SOLAR PANELS & BATTERY SYSTEM

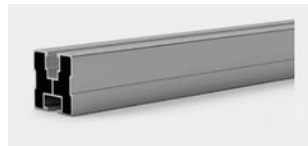
MOUNTING & COMPONENTS

THE DETAILS THAT MAKE THE INSTALLATION.

Original ProMount product images from the TLS-Solutions material proposal. Final dimensions, fixing spacing, ballast and article references are specified for the installation.



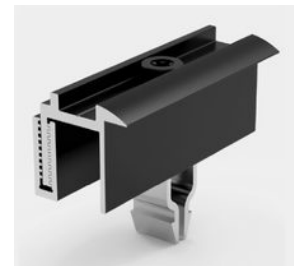
1. Double-adjustable roof hook



2. Aluminium mounting rail



3. Click middle clamp



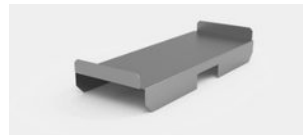
4. Click end clamp



5. SOLARSPEED east-west profile



6. SOLARSPEED side plate



7. Ballast holder

SELECTED EQUIPMENT. A CLEARLY DEFINED INSTALLATION.

Bauer glass-glass panels

465 Wp modules form the basis of our example packages. Bauer publishes a 30-year product and performance warranty, subject to the applicable model and conditions.

Solis inverter & monitoring

The inverter is matched to the supply and panel layout. Monitoring is specified with the appropriate communication equipment and internet connection.

Marstek energy storage

An optional AC-coupled storage route. Storage capacity, usable power and backup arrangements are specified for the selected model.

Design the support system for the roof structure and wind exposure. Preserve waterproofing, drainage and maintenance access. Confirm equipment, protections, metering and registration with the appointed electrical specialist.



SOLAR PANELS & BATTERY SYSTEM

APPLICATIONS · ENERGY & ELECTRICAL DESIGN

THREE SETTINGS. ONE CAREFULLY DESIGNED SYSTEM.



Pitched roofs



Flat roofs



Ground-mounted systems

Illustrative installation concepts; not completed EcoViva projects.

FROM SUNLIGHT TO USEFUL ELECTRICITY

1. Modules & DC circuit

Bauer 465 Wp glass-glass modules. Layout, string voltage and current must suit the selected inverter. Connector compatibility and cable protection are checked.

2. Inverter

A suitable Solis inverter converts DC generation to AC. The exact model, phase configuration, power and communication accessories are confirmed for the installation.

3. Protection, metering & distribution

DC and AC protection, earthing, cable sizes and metering follow the installation design. Connection and certification are performed by the appointed authorised specialist.

4. Household consumption

Electricity is used by the property while it is produced. Useful system size depends on the consumption profile and occupancy.

5. Grid interface

Import, export and surplus compensation depend on the selected self-consumption arrangement, registration and electricity contract.

6. Optional AC-coupled storage

Marstek VENUS E is an AC-coupled storage option. Version, controls and metering are confirmed; whole-house backup is not assumed.

Discuss your solar installation

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